

Architecture: Payments Platform Overview

Meridian Commercial Bank - Internal Document

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Overview

The Payments Platform handles card authorizations, ACH batch transfers, and wire transfers for Meridian Commercial Bank. It is composed of four primary services: the Payments Gateway (PGW), the Card Authorization Service (CAS), the ACH Batch Processor, and the Wire Transfer Service (WTS).

Components

- Payments Gateway (PGW): Entry point for all card-present and card-not-present authorization requests from merchants and internal channels. Routes to the Card Network Adapter (CNA), which integrates with external card network APIs (Visa/Mastercard equivalents in this fictional environment).
- Card Authorization Service (CAS): Maintains card status, limits, and authorization holds. Backed by a primary/replica relational database with automated failover.
- ACH Batch Processor: Runs nightly batch jobs to submit outbound ACH payments (payroll, vendor disbursements, customer transfers) to the Fed ACH network, and ingest inbound ACH files.
- Wire Transfer Service (WTS): Handles real-time, high-value wire transfers with mandatory compliance screening (sanctions/AML checks) before submission to the correspondent banking network.

Data Flow (Card Authorization)

Merchant terminal -> PGW -> CAS (limit/status check) -> CNA -> external card network -> response propagated back through CNA -> PGW -> merchant terminal. Target end-to-end latency: under 800ms p99.

Data Flow (Wire Transfer)

Customer request (web/mobile/branch) -> WTS -> Compliance Screening Service (sanctions/AML) -> WTS -> correspondent bank network -> confirmation back to WTS -> customer notification.

Reliability Posture

- PGW and CAS run active-active across two regions; CAS uses a primary/replica database with automated failover (see RB-database-restore).
- WTS and the ACH Batch Processor run active-passive; WTS has no automated regional failover as of this writing (tracked as a platform gap).
- Availability SLA: CAS and PGW 99.95%; WTS 99.9%; ACH Batch Processor 99.5% (batch, not real-time).

Known Gaps / Active Work

- Idempotency key support is being added to WTS following INC-2026-02-27-004 (duplicate wire transfers).
- Capacity planning for the ACH Batch Processor is being revised to be calendar-aware following INC-2025-12-19-003 (year-end volume spike).
- CNA connection pool configuration has caused two incidents (INC-2025-08-14-001, INC-2026-05-11-005) and is being migrated to enforced IaC configuration.

Related Documents

- RB-payment-gateway-failover
- INC-2025-08-14-001, INC-2025-10-02-002, INC-2025-12-19-003, INC-2026-02-27-004, INC-2026-05-11-005